Cross Validation

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The Machine

- Cross-validation (CV) is a **repeated out-of-sample (OOS) test**
  1. Split history into multiple **train → test** segments
  2. Fit the model on each train segment
  3. Evaluate on the following test segment
  4. Summarize performance across splits
- In time series sequential order is key, random shuffling not possible

The Why

- **Generalization across regimes:** different periods impact models differently
- CV highlights robustness across regimes
- **C**hoose features & model hyperparameters using repeated Out-of-Sample(OOS) evidence
  - 80/20 split is not sufficiently robust although fast and simple
- Highlights variance and potential model instability across folds; not just the mean
- **Leakage discipline -** forces “train - uses past only, test - uses future only”

Core Setup

Let:

- H - number of forecast horizons
- K – number of folds (n_splits)

This implies fit one model type– loop through

- fit H separate models
- for each horizon h, you run K folds
- total fits = H × K
- each fold produces predictions for all timestamps in that fold's test block

```
H = 3                                # horizons: 1..H
tscv = TimeSeriesSplit(n_splits=5, gap=H)

preds = {}                           # horizon -> full-length OOS predictions

for h in range(1, H + 1):
    y_h = y.shift(-h)                # observed target
    yhat = pd.Series(np.nan, index=X.index) # out of sample predictions

    for train_idx, test_idx in tscv.split(X):
        m = clone(model)             # fresh model each fold
        m.fit(X.iloc[train_idx], y_h.iloc[train_idx])
        yhat.iloc[test_idx] = m.predict(X.iloc[test_idx])

    preds[h] = yhat                  # stitched preds for horizon h
```

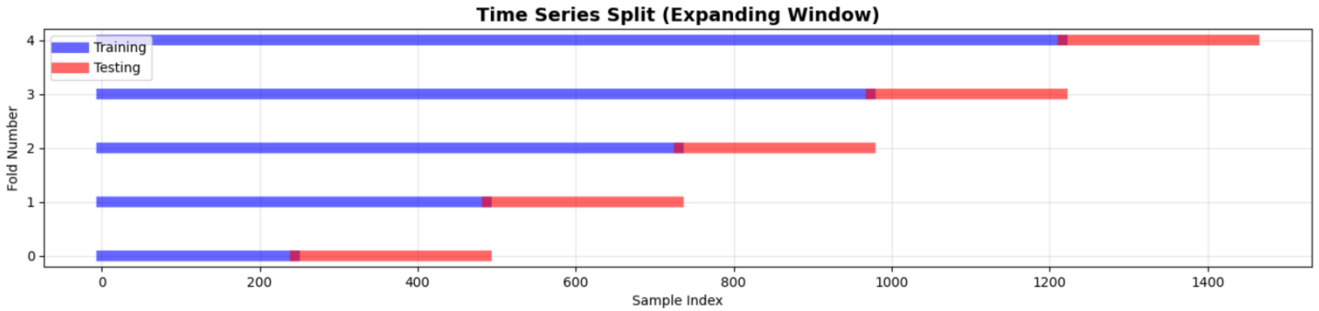
Walk Forward Cross Validation(CV)- Expanding Window vs. Rolling Window

CV exposes model performance dispersion across different periods since markets are subject to regime shifts

Time-series cross-validation - data is **split into folds -** split the timeline into a sequence of **repeated train & test** evaluations

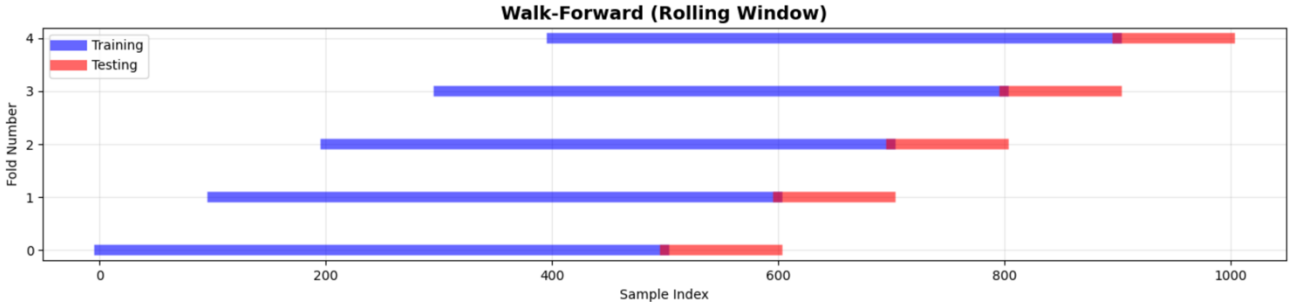
Setting up CV offer many sophisticated options, but first consider baseline Walk-Forward approach : `TimeSeriesSplit(… …)`

- **Walk-Forward** implies each fold trains on earlier data and tests on later data, then move forward and repeat
- Two Styles of Walk-Forward:
  - Expanding Window



- Training starts at the beginning and **grows** each fold
- Test is the next block
- Suitable when older data is still useful

  - Rolling window



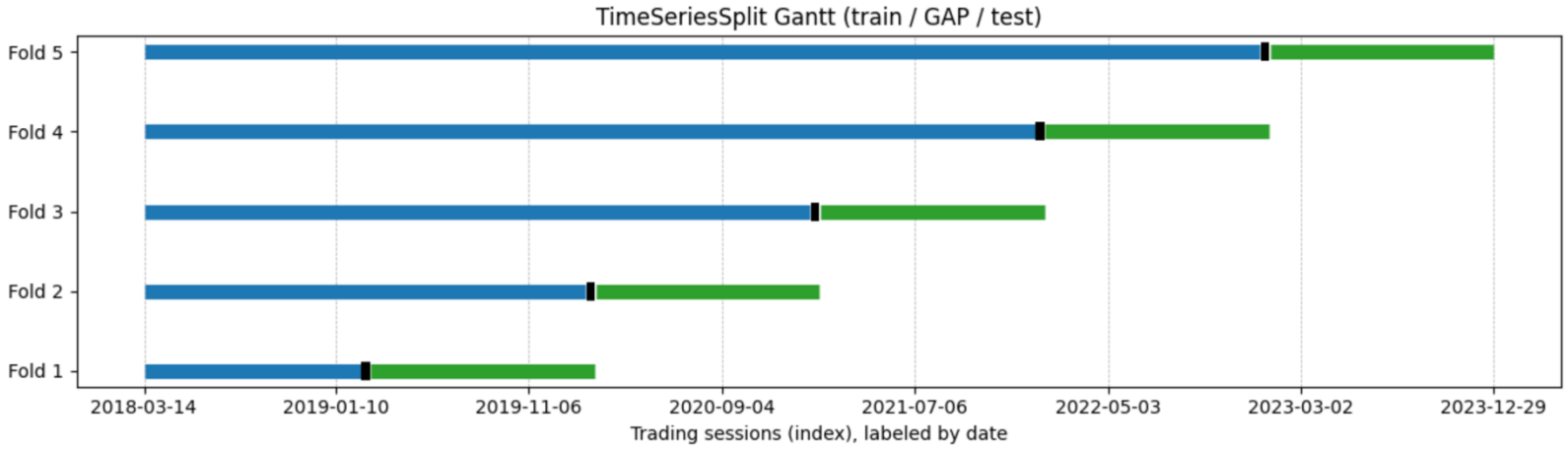
- **Training length is fixed** (a sliding window)
- **Each fold** uses only the **most recent *N* observations**
- Suitable when **drift dominates** and older regimes are less relevant

- Each fold gives you an **out-of-sample** performance measurement in a different time region
- We can measure **average performance** across folds

Walk Forward Cross Validation(CV) - Leakage Control

CV- Terminology

- **Labels** - labels are the **target values you want to predict**
- **Features** – self explanatory
- **Embargo** – "blank out interval" of observations excluded between training and test to prevent label/feature overlap
  - *tscv = TimeSeriesSplit(**n_splits=5, gap=H**)*
  - Embargo can be applied between Train and Test and between Test and the following Train
- **Purge -** remove any training rows that "peek into" the test window even if they aren't exactly at the boundary (variable horizons)
- Purge + Embargo –labels are irregular and Embargo alone is not enough prevent leakage



CV vs. “single 80/20 split” - Holdout Approach

Common approach:

- Train on first 80% of time, test on last 20% - single test
- Simple, fast, intuitive
- Mimics the “train on history → predict the future” story

Issues:

- **Performance** is assessed on a single period, can be misleading
- **Regime dependency -:** might be dominated by unusual (crisis, boom, low-vol), model might not generalize
- **Calibrating** hyperparameters observing the **same test set**, - the **“test” stops being truly out-of-sample**

Final Note - Multi-Horizon Predictions

Example :

- 1 - 3 days ahead prediction (daily time series), Simple LinearRegression

CV Walk Forward Approach:

$$r_{t+h} = \beta_0^{(h)}(t) + X_t^\top \beta^{(h)}(t) + \varepsilon_t^{(h)}$$

- Effectively we fit x3 different conditional expectation models $H_{horizon}$ x K_{Fold}
- Rerun regression for each horizon and get a different coefficient vector $\beta^{(h)}$
- Signal can differ by horizon
- Noise generally increases with horizon -
- Coefficient stability

Fixed-beta Time Invariant multi-horizon regression (a simplification):

$$\mathbb{E}[r_{t+h} \mid X_t] = \beta_0^{(h)} + X_t^\top \beta^{(h)}, \quad h \in \{1, ..., H\}, \quad \beta^{(h)} \text{ constant in } t$$

- Assumed **Linearity / additivity**
- Coefficient stability $\beta \; is \; constant$
- Error structure - classic OLS assumes ≈i.i.d. Errors, although errors are often heteroskedastic
- No regime dependence

80/20 Time Split - single holdout

- Train first 80% - test last 20% (one Out of sample window)
- One-shot estimate; can be regime-dependent (crisis/low-vol), and repeated tuning can overfit the holdout